A Neural Network Approach to Predicting Bike Sharing Trip Duration

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Introduction

- Bike sharing systems like Chicago's Divvy provide a healthy, affordable, and eco-friendly way to travel
- Thousands of daily trips create large amounts of data about how people move around the city
- Predicting trip duration can help improve bike availability, station planning, and overall system efficiency

Research Objectives

- Predict bike trip duration using historical data
- Develop a **Neural Network** model to learn patterns in trip behavior
- Compare the neural network against a simple baseline model
- Evaluate whether the Neural Network improves prediction accuracy

Methodology

- **Dataset:** 1 million Chicago Divvy bike trips, shuffled and split into training, validation, and test sets (consistent split for fair evaluation)
- **Key features:** Time of day, day of week, month, and trip distance
- **Models:** Linear Regression (baseline), Ridge and Lasso (variants of linear regression), and a Neural Network
- **Training:** Neural Network tuned using validation data; final results evaluated on a separate test set
- **Evaluation:** Performance measured using R^2 and MAPE to compare prediction accuracy

Results

- The **Neural Network achieved the best performance** across all evaluation metrics
- **R^2 (R-squared)** measures how well the model explains the data. Values closer to 1 mean better predictions
  - Neural Network: ~0.74
  - Linear, Ridge, Lasso: ~0.64
- **MAPE** (**Mean Absolute Percentage Error)** measures the average prediction error as a percentage. Lower values mean more accurate predictions
  - Neural Network: 22.36%
  - Linear, Ridge, Lasso: 29.61%
- All linear-based models (Linear Regression, Ridge, Lasso) performed nearly the same, while the Neural Network showed clear **improvement in both accuracy and error reduction**

Discussion

- The **Neural Network** performs better because it can **learn more complex patterns** in the data, such as how time and distance interact
- Linear Regression, Ridge, and Lasso are simpler models and cannot capture these complex relationships as well
- Ridge and Lasso are improved versions of Linear Regression that help prevent overfitting, but they still behave similarly in this problem
- The **improvement in accuracy shows** that trip duration depends on **multiple interacting factors**, not just simple relationships
- Neural Networks require **more training time** and **careful tuning** compared to simpler models

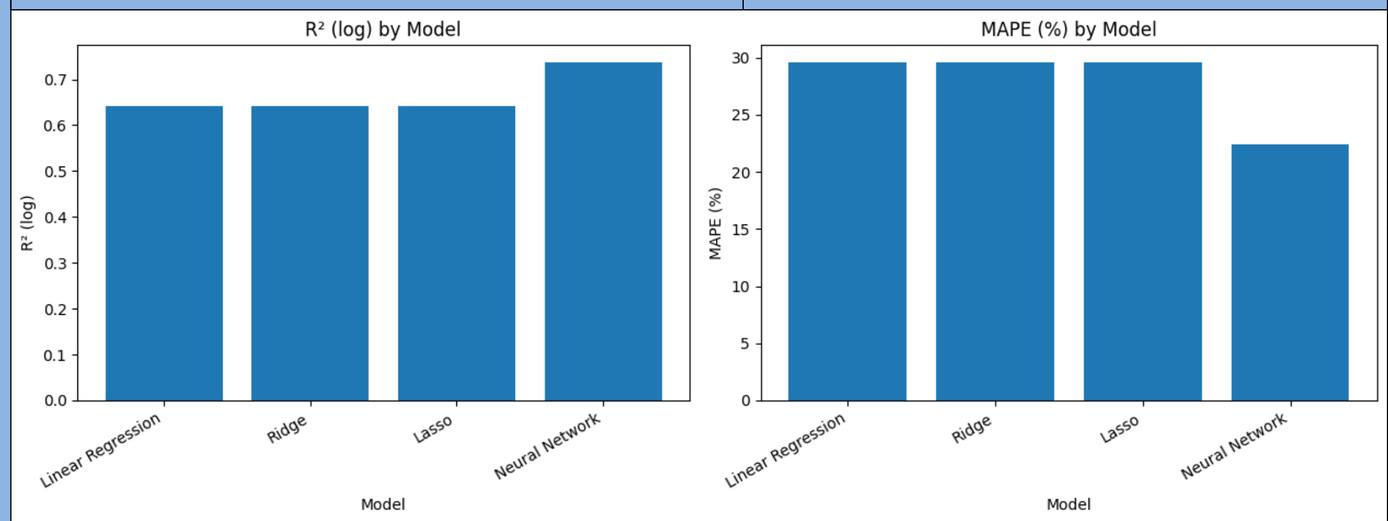

Figure 1. Comparison of model performance using R^2 (higher is better) and MAPE (lower is better). The Neural Network performs best, while Linear Regression, Ridge, and Lasso (closely related linear models) show similar results.

Conclusion

- Neural Networks provide more accurate predictions than traditional linear models for bike trip duration
- This improved accuracy can directly support
  - **Better bike availability**
  - **Smarter station placement**
  - **Improve user experience**
  - **Effective urban planning**
- While Neural Networks require more setup and training, their ability to capture real-world patterns makes them a **valuable tool for large, complex datasets**

References

1. **Kruse et al. (2022)** *Computational Intelligence: A Methodological Introduction*, Springer.
2. **Yingwurenjian (n.d)** "Chicago Divvy Bicycle Sharing Data", Kaggle. https://www.kaggle.com/datasets/yingwurenjian/chicago-divvy-bicycle-sharing-data